URBAN BOUNDARY LAYER

Academic English Vocabulary Test

城市边界层学术英语词汇自测卷

纯试卷版

15 道词汇选择｜ 1 篇完形填空｜ 2 篇阅读理解｜ 5 道语境填空

重点检测：词义辨析·固定搭配·语境推断·学术阅读

姓名		日期	
建议用时	60 分钟	满分	100 分
核心材料	*Progress in observing and modelling the urban boundary layer*		

论文语境核对。

作答说明

- 全卷共 40 题，满分 100 分。除第四部分外，每题只有一个最佳答案。
- 先独立完成试卷，再查看解析版。选择题请在题号旁写出 A、B、C 或 D。
- 阅读材料为原创文本；不要求具备城市气象学专业背景，所有答案均可由语言或文意推出。

PART Ⅰ Vocabulary & Collocation

30 分

Directions: Choose the option that best completes the sentence or explains the underlined expression. *(15 questions, 2 points each)*

1. 基础 Researchers have made _____ progress in observing the urban atmosphere.

A. considerate
B. considerable
C. considering
D. considered

2. 基础 _____ the limited number of long-term observations, the team avoided making a universal claim.

A. Giving
B. Given
C. Give
D. To giving

3. 基础 In the phrase "***sound*** modelling and observational tools," ***sound*** most nearly means:

A. producing a loud noise
B. financially profitable
C. reliable and well-founded
D. simple enough for beginners

4. 基础 If a need ***becomes pressing***, it:

A. becomes increasingly urgent
B. is gradually forgotten
C. becomes physically heavier
D. has already been satisfied

5. 中等 Which sentence uses ***in terms of*** most naturally?

A. In terms of the instrument broke, the team stopped.
B. The two methods differ in terms of spatial coverage.
C. The report is in terms of five pages.
D. The model in terms of predicts tomorrow.

6. 中等 A framework ***spanning the microscale to the mesoscale*** covers:

A. only processes smaller than a building
B. only regional weather systems
C. a range extending from very local to city-regional scales
D. two unrelated periods of historical time

7. 基础 In "the main determinants of boundary-layer structure," ***determinants*** are:

A. measuring devices
C. final conclusions
B. controlling factors
D. geometric boundaries

8. 中等 If assumptions about atmospheric turbulence ***break down*** near tall buildings, the assumptions:

A. become more detailed
B. are divided into smaller assumptions
C. cease to be valid in that setting
D. are translated into another language

9. 基础 Surface heterogeneity may ***dominate*** the observed pattern. This means it may:

A. have the strongest influence on the pattern
B. disappear from the pattern
C. measure the pattern remotely
D. make the pattern perfectly uniform

10. 中等 The ***coupling of the surface to the urban boundary layer*** refers to:

A. the physical interaction linking the surface and the air above it
B. the distance between two independent instruments
C. the boundary separating a city from the countryside
D. the pairing of two words with the same meaning

11. 基础 In this study, to ***model*** airflow means to:

A. present a fashionable example for other cities to copy
B. represent or simulate the physical process
C. draw only the administrative boundary of a city
D. build a small physical object and avoid all calculations

12. 进阶 The part of the lower atmosphere that responds directly to the city's surface is called the:

A. urban canopy
C. urban boundary layer
B. residual layer
D. high-rise boundary

13. 中等 ***Surface heterogeneity*** is best understood as:

A. spatial differences in materials, vegetation, shape, or heat release
B. a perfectly even surface extending across an entire city
C. an error caused only by a damaged surface sensor
D. the vertical distance from the road to a roof

14. 基础 A ***spatially extensive*** measurement network:

A. operates for one minute
C. uses only expensive sensors
B. covers a large geographic area
D. measures one variable in one room

15. 中等 Which sentence is grammatically correct and closest to the target meaning?

A. New tools permit researchers make rapid progress.
B. New tools permit to make researchers rapid progress.
C. New tools permit researchers to make rapid progress.
D. New tools permit researchers making rapidly progresses.

PART II Cloze

20 分

Directions: Read the passage and choose the best answer for each blank. *(10 questions, 2 points each)*

Building a Study Across Scales

Cities are difficult places in which to study the atmosphere. A research team starting a new project therefore needs a clear conceptual **(16)** ____. The plan should not stop at a single street: it may need to **(17)** ____ the microscale to the mesoscale.

At street level, the **(18)** ____ formed by buildings and roads creates complex flows. In districts filled with **(19)** ____ buildings, wakes from individual structures can reach far above roof level. As a result, simple assumptions about turbulence may **(20)** ____ in the urban roughness sub-layer.

The challenge changes again at larger scales. Differences among parks, concrete blocks, water, and industrial land create surface **(21)** ____. Under some weather conditions, these contrasts can **(22)** ____ the pattern recorded by a small number of stations. Night-time work is especially valuable because the **(23)** ____ above the shallow near-surface layer still remains under-researched.

To connect all these scales, the team combines **(24)** ____, including lidar, with **(25)** ____ that represent streets and neighbourhoods in greater detail. The observations are collected by a spatially extensive network and are coupled to model development. Together, these choices make the conclusions more defensible.

16. A. framework
B. determinant
C. boundary
D. priority

17. A. press
B. span
C. modelled
D. couple

18. A. residual layer
B. urban canopy
C. remote method
D. research priority

19. A. high-rise
B. long-term
C. spatially
D. conceptual

20. A. make up
B. break down
C. carry out
D. turn over

21. A. heterogeneity
B. boundary
C. progress
D. pressure

22.

A. develop
B. permit
C. dominate
D. remain

23. A. urban canopy
B. residual layer
C. microscale
D. model boundary

24. A. remote sensing methods
B. direct touching methods
C. conceptual priorities
D. high-rise assumptions

25. A. highly resolved model tools
B. high-resolution modelling tools
C. modelling high-resolution tools
D. high resolutions modelling tool

PART III Reading Comprehension

30 分

Directions: Read the two passages and choose the best answer. *(10 questions, 3 points each)*

Passage A —When One Neighbourhood Is Not Enough

Maya began her project with a weather tower in a compact city centre. Her first model treated the city as a uniform surface, and its equations worked reasonably well above open ground. Close to the buildings, however, the model repeatedly underestimated sudden changes in wind. The problem was not simply bad data. The tower stood inside the urban roughness sub-layer, where wakes from roofs and walls disrupted the flow. Assumptions developed for a flatter surface could break down there.

At first, Maya tried to improve the model by tuning one parameter. This helped on winds from the west but made the results worse on winds from the south. She then mapped the tower's surroundings and found parks, glass towers, brick blocks, and a river within a few kilometres. Their different shapes and thermal properties created strong surface heterogeneity. No single adjustment could represent every wind direction. The finding showed that surface differences should be more strongly considered rather than averaged away.

Maya redesigned the project in terms of a scale-aware conceptual framework. Street-level sensors examined individual wakes, lidar followed the atmosphere above the roofs, and several stations sampled different neighbourhoods. High-resolution modelling tools were used to connect these observations. The new design spanned the microscale to the mesoscale instead of treating one tower as a miniature version of the whole city.

The redesign did not make the city simple. It made the limits of each measurement explicit. That change was scientifically valuable: a sound study does not require every instrument to represent everything. It requires the researcher to know what each instrument can and cannot represent.

26. What is the main purpose of the passage?

A. To argue that all urban weather towers should be removed

B. To show why an urban study had to account for different scales and surfaces

C. To prove that lidar always gives better data than a weather tower

D. To explain how one parameter can represent every neighbourhood

27. Why did the first model perform poorly close to the buildings?

A. Its assumptions were not fully valid in the roughness sub-layer.

B. The tower measured temperature instead of wind.

C. The city contained no open ground at all.

D. The model was designed only for night-time conditions.

28. In the second paragraph, "should be more strongly considered" means that surface differences should:

A. receive greater attention in the study design

B. be measured only with physically stronger instruments

C. be removed before any model is tested

D. be assumed to have the same effect in every wind direction

29. The phrase "a miniature version of the whole city" is used to reject the idea that:

A. one site automatically represents all urban environments

B. city-scale models may include local observations

C. instruments have limits that should be reported

D. street-level sensors can examine individual wakes

30. Which statement would the author most likely agree with?

A. A sound study makes the scope and limits of its evidence clear.

B. More instruments automatically remove every source of uncertainty.

C. A uniform-surface model is accurate in every high-rise district.

D. Measurements are useful only when they cover the entire city.

Passage B —Studying the City After Sunset

During the day, heat from the surface helps mix air through a deep part of the urban boundary layer. After sunset, that continuous heating weakens. A shallow, more stable layer may then form near the ground, while some of the formerly mixed air remains above it. This upper air is called the residual layer. It can retain traces of the day's heat, moisture, and pollution, so it may influence conditions the following morning.

Despite that possible influence, the residual layer remains under-researched. Many towers do not extend high enough to sample it, and balloon observations are too infrequent to describe its changes throughout every night. Remote sensing methods can fill part of the gap by providing repeated vertical profiles. Yet a single signal can be misleading. For example, the top of an aerosol-rich layer is not always the same as the top of active turbulence. A sound interpretation therefore compares several kinds of evidence rather than treating one detected boundary as certain.

Researchers are now developing spatially extensive networks in which lidars, surface stations, and occasional balloons operate together. The observations can be coupled to high-resolution models, allowing

teams to test whether the same explanation works on clear, cloudy, calm, and windy nights. This approach also helps identify specific research priorities instead of collecting data without a clear question. Understanding the night-time atmosphere has become pressing as cities seek to prepare for heat and poor air quality. Better evidence will not by itself develop cities sustainably, but it can give planners a more reliable basis for choosing and evaluating actions.

31. According to the passage, what is the residual layer?

A. A shallow stable layer that always touches the ground
B. Formerly mixed daytime air remaining above a new near-surface layer
C. The solid material left on rooftops after pollution events
D. A permanent boundary separating a city from the countryside

32. Why has the residual layer remained under-researched?

A. It exists only during unusual storms.
B. Its temperature cannot be represented by any model.
C. Common observations often lack sufficient height or time coverage.
D. Researchers have already agreed that it has no practical influence.

33. In the second paragraph, ***sound interpretation*** most nearly means:

A. an audible explanation
B. a reliable interpretation
C. a brief description
D. a popular opinion

34. Which research design does the author support?

A. Using one aerosol boundary as certain evidence of turbulence
B. Collecting more data before deciding on any research question
C. Combining several observation types with high-resolution models
D. Replacing all observations with a single city-wide model

35. What can reasonably be inferred from the final paragraph?

A. Scientific evidence can support planning, but it does not make policy choices by itself.
B. Sustainable urban development depends only on night-time temperature.
C. Once a city installs lidar, poor air quality will disappear.
D. Urban planners should wait until every uncertainty has been eliminated.

PART IV Contextual Completion

20 分

Directions: Complete each sentence with the exact English expression suggested by the Chinese cue. You may change capitalization, but do not change the core wording. *(5 questions, 4 points each)*

36. Given rapid urban growth, learning how to 以可持续方式发展城市 has become pressing.

_____

37. 本综述考察 how street design changes wind and temperature.

_____

38. The final chapter summarizes the 在夜间城市热研究中取得的进展.

_____

39. The evidence is organized 依据一个跨越微尺度到中尺度的概念框架.

_____

40. For a trustworthy result, a high-resolution model 必须与观测相耦合 rather than be used alone.

_____

交卷前自检 检查 *sustainably* 是否正确修饰动词；检查 *progress* 是否作不可数名词；检查 *in terms of* 与 *be coupled to* 的完整结构。